

Is the Canadian Home Bias Mean-Variance Efficient?

Constructing the Most Efficient Portfolio

CFM301 Project 1 | Jian | September 2026

1. Economic Goal & Motivation

Canadian investors display one of the world's strongest equity home biases: they allocate roughly 50% of equity portfolios to Canadian stocks (IMF Coordinated Portfolio Investment Survey, via Vanguard, 2024), even though Canada is only about 3% of global market capitalization -- a 15-18x overweight. Vanguard's own minimum-variance analysis recommends just 30% domestic equity for Canadians. This project asks: is the observed home bias mean-variance efficient? Tracing the long-only frontier over regional equity markets in Canadian dollars, we compute the domestic weight a mean-variance optimizer chooses and compare it to (i) the ~50% Canadians actually hold and (ii) Canada's ~3% market-cap weight.

2. Data

Source. Monthly total returns of five regional equity ETFs from Yahoo Finance (adjusted closes, which reinvest distributions): XIU.TO (Canada, S&P/TSX 60), SPY (US large cap), IWM (US small cap), EFA (developed ex-NA), EEM (emerging markets). The brief specifies CRSP/Compustat via WRDS; this environment has no WRDS access, so listed-ETF total returns serve as the reproducible proxy. The pipeline reads only `data/monthly_returns_cad.csv`, so a WRDS/CRSP country-index extract can be substituted without changing any downstream code.

Sample & cleaning. Month-end bars, January 2003-August 2026. USD-denominated ETFs are converted to CAD each month-end via the USD/CAD spot rate ($CAD=X$): $R_{CAD} = (1+R_{USD})(1+R_{FX})-1$. Simple returns are computed from adjusted closes; the incomplete September 2026 bar is dropped; the common sample is October 2003-August 2026 (275 months). Means are annualized $\times 12$, covariances $\times 12$ (volatilities $\times \sqrt{12}$).

Roadmap. Section 3 reports the estimated means, volatilities and correlations; Section 4 traces the frontier, states the selection criterion and gives the optimal weights; Section 5 plots the frontier with the chosen portfolio marked; Section 6 interprets the results.

3. Estimated Inputs (annualized, CAD)

	Mean	Stdev
Canada	0.1056	0.1276
US Large Cap	0.1249	0.1691
US Small Cap	0.1185	0.2156
Dev ex-NA	0.0928	0.1839
Emerging	0.1124	0.2225

Correlation matrix (monthly returns)

	Canada	US Large Cap	US Small Cap	Dev ex-NA	Emerging
Canada	1.00	0.67	0.69	0.69	0.68
US Large Cap	0.67	1.00	0.90	0.89	0.77
US Small Cap	0.69	0.90	1.00	0.81	0.74
Dev ex-NA	0.69	0.89	0.81	1.00	0.87
Emerging	0.68	0.77	0.74	0.87	1.00

4. Frontier, Selection Criterion & Optimal Portfolio

Frontier. For target returns m^* spanning the asset means, we solve $\min w'Sw$ s.t. $w'\mu = m^*$, $\sum(w) = 1$, $w \geq 0$ (SLSQP quadratic programs). Only the upper branch is efficient. Long-only is imposed because the question concerns the strategic allocation of a representative retail investor, for whom shorting country indices is impractical; it also keeps weights interpretable as portfolio shares.

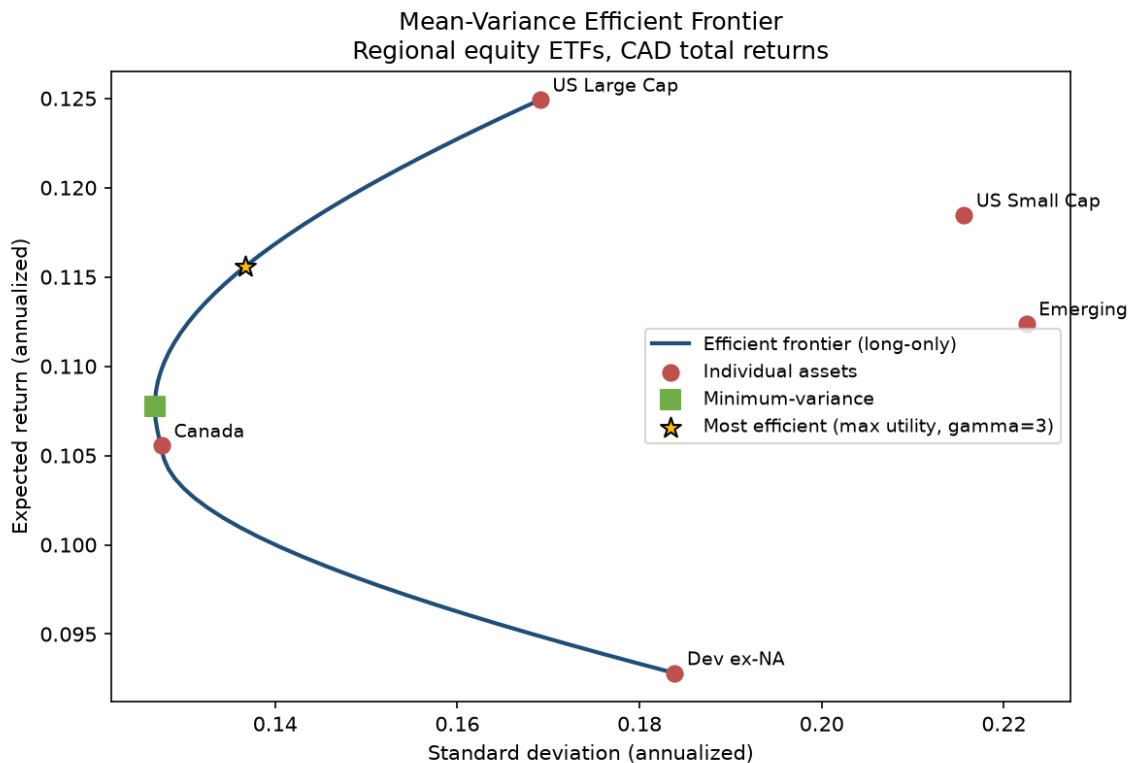
Selection criterion. With no risk-free asset there is no tangency/Sharpe criterion, so we maximize mean-variance utility $U = E[rp] - (g/2)Var[rp]$ with $g = 3$, the course convention (Ang, Ch. 3: typical risk aversion 1-10; the lecture's own G5 example uses $g = 3$ with no shorting).

	Min-Variance	Max utility (g=3)
Canada	88.85%	48.17%
US Large Cap	11.15%	51.83%
US Small Cap	0.00%	0.00%
Dev ex-NA	0.00%	0.00%
Emerging	0.00%	0.00%

Most efficient portfolio: 48.2% Canada / 51.8% US large cap, $E[r] = 11.56\%$, $\sigma = 13.68\%$. (Minimum-variance for reference: 88.9% Canada / 11.1% US large cap, $E[r] = 10.77\%$, $\sigma = 12.68\%$.) Weights sum to 1 and satisfy the long-only constraint. Sensitivity to risk aversion:

	$E[r]$	σ	Canada	US LC	US SC	EAFE	EM
1	0.12	0.17	0.00	1.00	0.00	0.00	0.00
2	0.12	0.15	0.28	0.72	0.00	0.00	0.00
3	0.12	0.14	0.48	0.52	0.00	0.00	0.00
5	0.11	0.13	0.64	0.36	0.00	0.00	0.00
10	0.11	0.13	0.77	0.23	0.00	0.00	0.00

5. Frontier Plot



6. Economic Conclusion

The optimizer's answer is striking: at $g = 3$ the most efficient portfolio holds 48% Canadian equity -- essentially the ~50% home bias Canadians actually exhibit. Across the plausible risk-aversion range the optimal domestic weight runs from 28% ($g = 2$) to 64% ($g = 5$), bracketing both the observed 50% and Vanguard's 30% recommendation (which corresponds to about $g = 4$ -5 here). In other words, the Canadian home bias is roughly mean-variance efficient for a moderately risk-averse investor -- it is not the glaring inefficiency its 15-18x market-cap overweight suggests. The intuition: Canada had the lowest volatility (12.8%) and the highest mean among the diversifiers once correlations (~0.7-0.9) are accounted for.

Three caveats. First, the solution is a corner: US small cap, EAFE and emerging markets receive zero weight at every risk-aversion level -- the textbook 'error maximization' problem (Michaud, 1989; Ang, Ch. 3), where small input differences produce extreme, concentrated portfolios. Second, the minimum-variance portfolio would imply 89% domestic equity, far more home bias than observed, so the 'efficient' domestic weight hinges on the criterion chosen. Third, inputs are sample estimates over 2003-2026; currency is unhedged and fees/taxes (e.g., the Canadian dividend tax credit) are ignored, all of which favor domestic holdings in practice.

Peer evaluation (three differences vs. paired group's solution): to be completed once the paired group is assigned.

References & Packages

- Ang, A. (2014). Asset Management, Ch. 3: Mean-Variance Investing (lecture slides).
- Michaud, R. (1989). The Markowitz optimization enigma. Financial Analysts Journal.
- Vanguard Canada home-bias research (2024), via Investment Executive: Canadians hold ~50% domestic equity vs ~3% world market cap; 30% recommended.
- Python: numpy, pandas, scipy (SLSQP), matplotlib, yfinance; report: fpdf2.